Differential and Integral Calculus

for Computer Scientists

Lecture notes with derivations, examples, figures, and exercises

These notes present single-variable and multivariable calculus as a mathematical toolkit for computer science: asymptotic reasoning, numerical computation, optimization, machine learning, graphics, data analysis, and algorithms.

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How to read these notes

Calculus is often introduced through motion, forces, and physical intuition. This text takes a different route. The main motivation is computation: functions as data transformations, derivatives as local linear approximations, integrals as accumulation, and gradients as the language of optimization.

The notes are written for a mathematics lecture addressed to computer scientists. They are not programming notes, but they constantly point to the mathematical structures that appear in algorithms, numerical methods, data science, graphics, and machine learning.

Computer science lens: what is essential for a computer scientist?

A computer scientist should be able to:

1. reason about limiting behavior and growth rates;
2. recognize when a function is continuous or differentiable;
3. use derivatives for approximation, sensitivity analysis, and optimization;
4. understand integrals as limits of sums;
5. use numerical integration with error awareness;
6. work with gradients, Jacobians, and Hessians in multivariable problems.

1 Functions and the computational point of view

1.1 Functions as mathematical models of computation

A function is a rule assigning to each input from a domain exactly one output. In computer science this idea appears everywhere: a program maps input data to output data, a loss function maps parameters to an error value, a hash function maps objects to addresses, and an activation function maps a real number to another real number.

Definition: function

A function from a set A to a set B, denoted

$$f : A \to B,$$

assigns to every element $x \in A$ exactly one element $f(x) \in B$. The set A is called the *domain*; the set B is called the *codomain*. The set

$$f(A) = \{f(x) : x \in A\}$$

is called the *image* or *range* of f.

In calculus we most often study functions

$$f : \mathbb{R} \to \mathbb{R}$$

or, later,

$$f : \mathbb{R}^n \to \mathbb{R}^m.$$

The first case describes one real input and one real output. The second case describes vector-valued data transformations.

1.2 Elementary families of functions

A useful calculus toolbox contains the following function families:

Family	Example	Typical role in CS
Polynomial	$x^3 - 2x + 1$	approximation, interpolation, algorithms
Rational	$\frac{x^2+1}{x-3}$	complexity formulas, transforms
Exponential	$e^x, 2^x$	growth, probabilities, algorithms
Logarithmic	$\log x$	complexity, information, entropy
Trigonometric	$\sin x, \cos x$	signals, graphics, Fourier methods
Piecewise	$\max(0, x)$	activations, thresholds, branching

Computer science lens: why continuous mathematics appears in discrete computing

Computer programs run on discrete machines, but many objects studied in computer science are naturally approximated by continuous models: probability densities, optimization landscapes, embeddings, image intensities, differentiable loss functions, splines, and numerical simulations. Calculus gives a language for these models and for the algorithms operating on them.

2 Limits

2.1 Intuitive meaning of a limit

A limit describes the behavior of a function near a point, not necessarily at that point. We write

$$\lim_{x \to a} f(x) = L$$

when the values $f(x)$ become arbitrarily close to L whenever x is sufficiently close to a, with $x \neq a$.

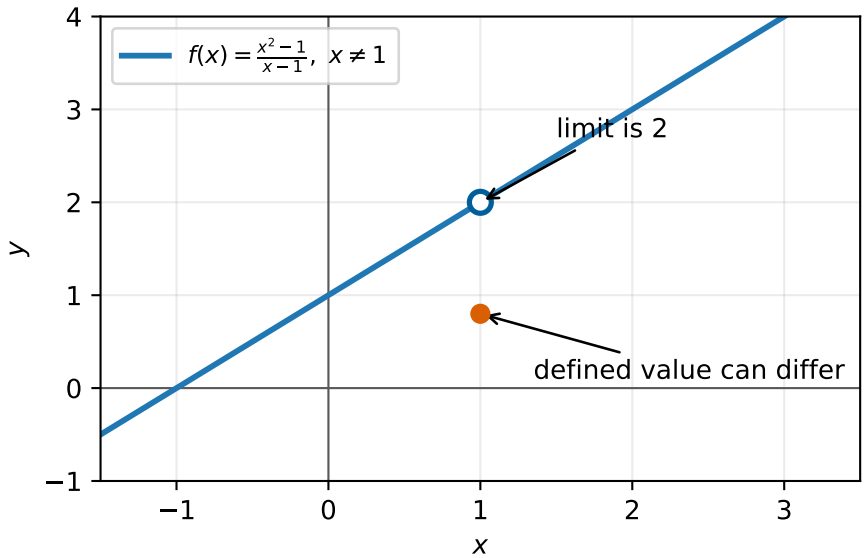

Figure 1: A function can have a limit at a point even if its value at that point is different or not defined.

Definition: limit at a point

Let f be defined in a punctured neighborhood of a. We say that

$$\lim_{x\to a} f(x) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x-a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The condition $0 < |x-a|$ excludes the point a itself. A limit is about what happens when we approach a, not about the value at a.

2.2 One-sided limits

For functions defined on intervals, we often need one-sided limits:

$$\lim_{x\to a^-} f(x), \qquad \lim_{x\to a^+} f(x).$$

The ordinary two-sided limit exists if and only if both one-sided limits exist and are equal.

Example: a jump prevents a two-sided limit

Consider

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then

$$\lim_{x\to 0^-} f(x) = 0, \qquad \lim_{x\to 0^+} f(x) = 1.$$

Since the one-sided limits are different, $\lim_{x\to 0} f(x)$ does not exist.

2.3 Limit laws

If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$, then:

$$\begin{aligned} \lim_{x\to a}(f(x) + g(x)) &= L + M, \\ \lim_{x\to a}(f(x)g(x)) &= LM, \\ \lim_{x\to a} \frac{f(x)}{g(x)} &= \frac{L}{M}, \quad M \neq 0. \end{aligned}$$

These rules allow us to compute many limits by substitution, as long as no indeterminate expression appears.

Example: removable singularity

For $x \neq 1$,

$$\frac{x^2-1}{x-1} = x + 1.$$

Therefore

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1}(x+1) = 2.$$

The original expression is undefined at $x = 1$, but the limit exists.

2.4 Important standard limits

The following limits are basic tools:

$$\lim_{x\to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x\to 0} \frac{1-\cos x}{x} = 0,$$

$$\lim_{x\to 0} \frac{e^x - 1}{x} = 1, \qquad \lim_{x\to 0} \frac{\log(1+x)}{x} = 1.$$

They are used in differentiation, Taylor expansions, numerical approximations, and asymptotic analysis.

3 Limits and asymptotic notation

3.1 Growth comparison through limits

Computer science often asks not for the exact value of a function, but for its growth rate. Limits provide a precise language for this.

Definition: asymptotic comparison

Let f, g be positive for sufficiently large x. We write

$$f(x) \sim g(x) \quad (x \to \infty)$$

if

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 1.$$

We say that f and g are asymptotically equivalent.

Example: dominant terms

For

$$f(n) = 3n^2 + 5n \log n + 100,$$

we have

$$\lim_{n\to\infty} \frac{f(n)}{3n^2} = 1.$$

Thus $f(n) \sim 3n^2$. The quadratic term dominates.

3.2 Big-O, little-o, and Theta

Definition: Big-O and little-o

Let f, g be functions defined for sufficiently large x, with $g(x) > 0$. We write:

- $f(x) = \mathcal{O}(g(x))$ if there exist $C > 0$ and x_0 such that $|f(x)| \le Cg(x)$ for all $x \ge x_0$;
- $f(x) = o(g(x))$ if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$;
- $f(x) = \Theta(g(x))$ if $f(x) = \mathcal{O}(g(x))$ and $g(x) = \mathcal{O}(f(x))$.

Computer science lens: why limits matter in algorithm analysis

A statement such as

$$n \log n = o(n^{1.1})$$

means that for large inputs, $n^{1.1}$ eventually dominates $n \log n$, even though for realistic finite values the comparison may still depend on constants. Limits describe eventual behavior, not necessarily practical runtime on a particular machine.

4 Continuity

4.1 Definition and intuition

Continuity means that small changes in input produce small changes in output. In computation, this is linked to stability: if input data is perturbed slightly, a continuous model does not jump unpredictably.

Definition: continuity at a point

A function f is continuous at a if:

1. $f(a)$ is defined;
2. $\lim_{x \to a} f(x)$ exists;
3. $\lim_{x \to a} f(x) = f(a)$.

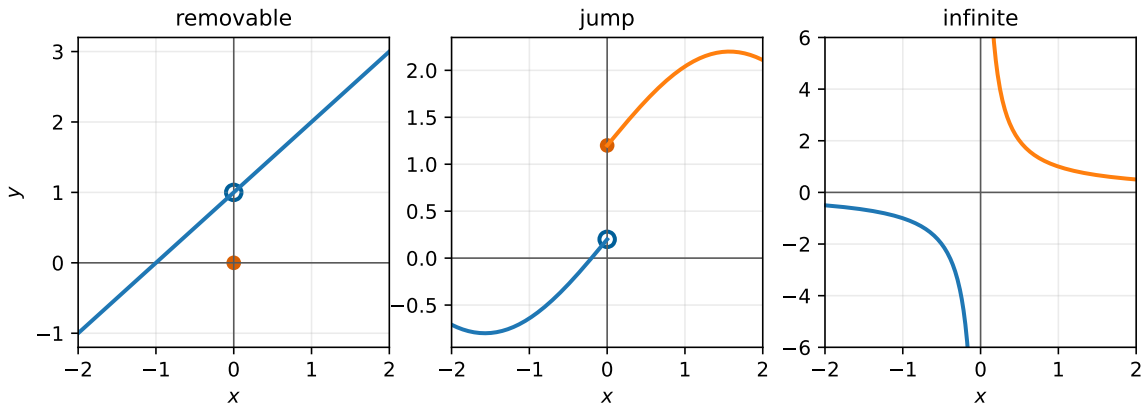

Figure 2: Three typical types of discontinuity: removable, jump, and infinite.

4.2 Continuity on intervals

A function is continuous on an interval if it is continuous at every point of that interval. Polynomials are continuous everywhere. Rational functions are continuous wherever the denominator is not zero. The functions e^x, $\log x$, $\sin x$, and $\cos x$ are continuous on their domains.

4.3 Intermediate value theorem

Theorem: intermediate value theorem

If f is continuous on $[a, b]$ and y lies between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that

$$f(c) = y.$$

This theorem guarantees existence. It does not necessarily give a formula for c. In computation, this distinction is important: existence and construction are different matters.

Computer science lens: bisection method

If f is continuous and $f(a)f(b) < 0$, then the intermediate value theorem guarantees a root in (a, b). The bisection method repeatedly halves the interval containing the sign change. After k iterations, the interval length is

$$\frac{b-a}{2^k}.$$

Thus the error decreases exponentially in the number of iterations.

4.4 Extreme value theorem

Theorem: extreme value theorem

If f is continuous on a closed and bounded interval $[a, b]$, then f attains a maximum and a minimum on $[a, b]$.

The hypotheses matter. On an open interval a continuous function need not attain a maximum or minimum. For example, $f(x) = x$ on $(0, 1)$ has neither a maximum nor a minimum.

5 Derivatives

5.1 Derivative as a limit

The derivative measures the best linear approximation of a function near a point. It starts from the slope of a secant line:

$$\frac{f(a+h) - f(a)}{h}.$$

As $h \to 0$, the second point approaches the first and the secant line approaches the tangent line, if the limit exists.

Definition: derivative

The derivative of f at a is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists.

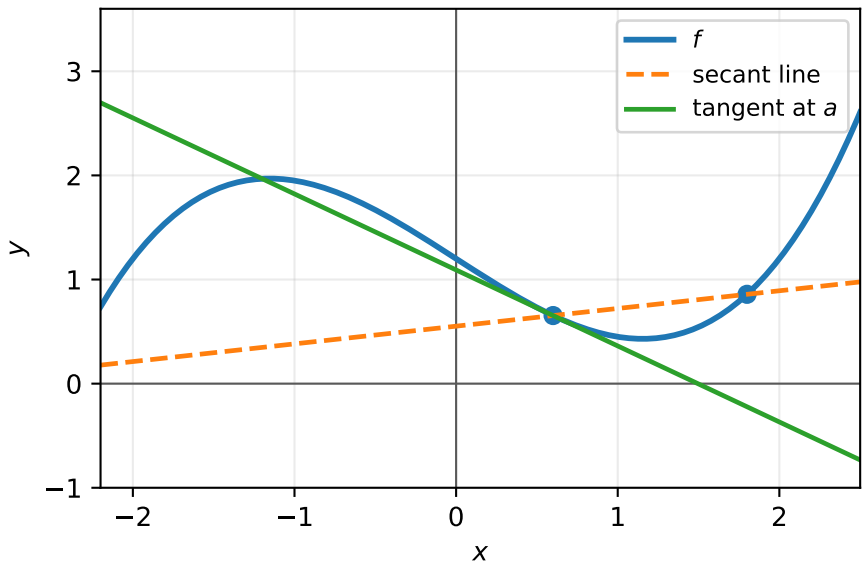

Figure 3: The derivative is obtained as the limiting slope of secant lines.

5.2 Local linear approximation

If f is differentiable at a, then for small h,

$$f(a+h) \approx f(a) + f'(a)h.$$

Equivalently,

$$f(x) \approx f(a) + f'(a)(x-a).$$

This is the equation of the tangent line.

Computer science lens: sensitivity analysis

If $f(x)$ is the cost of a computation depending on a parameter x, then $f'(a)$ approximates how sensitive the cost is to a small change near a:

$$\Delta f \approx f'(a)\Delta x.$$

This idea appears in numerical analysis, optimization, parameter tuning, and differentiable programming.

5.3 Differentiability implies continuity

Theorem: differentiability implies continuity

If f is differentiable at a, then f is continuous at a.

Proof. We write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a), \qquad x \neq a.$$

Taking $x \to a$, the quotient tends to $f'(a)$, while $x - a \to 0$. Hence

$$\lim_{x \to a}(f(x) - f(a)) = f'(a) \cdot 0 = 0.$$

Therefore $\lim_{x \to a} f(x) = f(a)$, so f is continuous at a. □

The converse is false: a continuous function may fail to be differentiable.

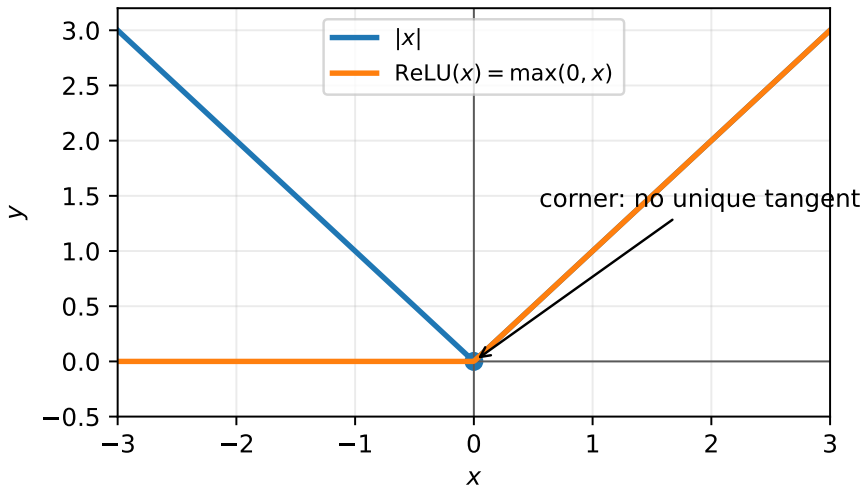

Figure 4: The functions $|x|$ and ReLU are continuous but not differentiable at 0.

6 Differentiation rules

6.1 Basic rules

If c is a constant, then:

$$\frac{d}{dx}c = 0, \qquad \frac{d}{dx}x^n = nx^{n-1}.$$

For differentiable functions f, g:

$$(f+g)' = f' + g', \qquad (cf)' = cf'.$$

6.2 Product, quotient, and chain rules

Theorem: product rule

If f and g are differentiable, then

$$(fg)' = f'g + fg'.$$

Theorem: quotient rule

If f and g are differentiable and $g(x) \neq 0$, then

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Theorem: chain rule

If g is differentiable at x, and f is differentiable at $g(x)$, then

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

Computer science lens: chain rule and computational graphs

The chain rule is the mathematical foundation of backpropagation. A complex function is represented as a composition of simpler functions. Derivatives are propagated through the

composition by repeated multiplication of local derivatives.

6.3 Derivatives of standard functions

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \log a$
$\log x$	$1/x$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$1/\cos^2 x$

6.4 Logarithmic differentiation

For positive functions, logarithmic differentiation can simplify products and powers. Suppose

$$y = x^x, \qquad x > 0.$$

Then

$$\log y = x \log x.$$

Differentiating both sides gives

$$\frac{y'}{y} = \log x + 1.$$

Therefore

$$y' = x^x(\log x + 1).$$

7 Functions frequently used in computer science

7.1 Logistic sigmoid

The logistic sigmoid is

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

It maps $\mathbb{R}$ to $(0, 1)$ and is often used to transform a real score into a value interpretable as a probability.

A useful identity is

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)).$$

7.2 ReLU and non-smooth models

The rectified linear unit is

$$\mathrm{ReLU}(x) = \max(0, x).$$

It is continuous everywhere, differentiable except at 0, and has derivative

$$\mathrm{ReLU}'(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases}$$

At 0, the classical derivative does not exist.

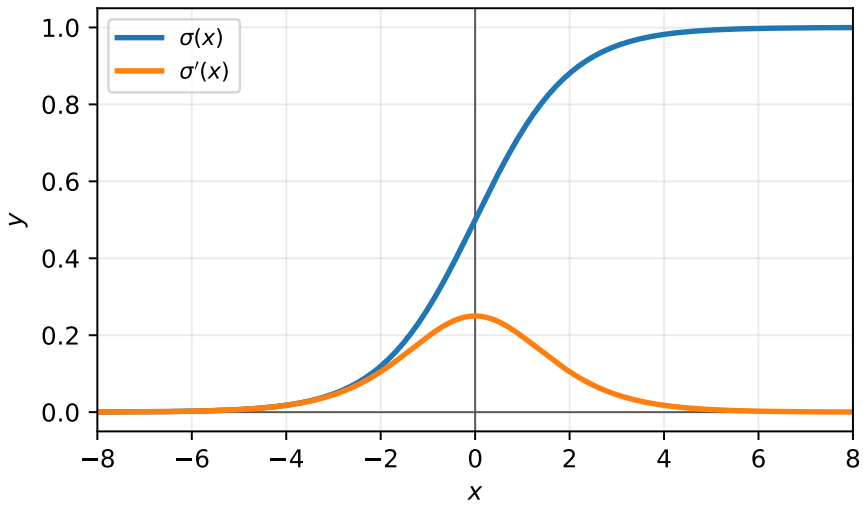

Figure 5: The logistic sigmoid and its derivative. The derivative is largest near zero and small in saturated regions.

Warning: differentiability in machine learning

Many optimization algorithms are described using gradients, but practical models may include non-differentiable points. This is handled using subgradients, smoothing, or by observing that the problematic points form a small set in many applications. Classical calculus still provides the baseline intuition.

7.3 Softplus as a smooth approximation

The softplus function is

$$s(x) = \log(1 + e^x).$$

Its derivative is the sigmoid:

$$s'(x) = \frac{e^x}{1 + e^x} = \sigma(x).$$

Softplus is a differentiable approximation of ReLU.

8 Higher derivatives and shape of functions

8.1 Second derivative

The second derivative is the derivative of the derivative:

$$f''(x) = (f'(x))'.$$

It describes how the slope changes.

- If $f''(x) > 0$ on an interval, the graph is convex there.
- If $f''(x) < 0$ on an interval, the graph is concave there.
- Points where concavity changes are candidates for inflection points.

8.2 Convexity

Definition: convex function on an interval

A function $f : I \to \mathbb{R}$ is convex if for all $x, y \in I$ and all $t \in [0, 1]$,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y).$$

For twice differentiable functions on an interval,

$$f''(x) \geq 0 \quad \text{for all } x$$

implies that f is convex.

Computer science lens: convex loss functions

Convexity is central in optimization. If a differentiable convex function has a critical point, that point is a global minimizer. This is why convex models are mathematically attractive: local information can certify global optimality.

9 Taylor approximation

9.1 From tangent lines to polynomials

The first-order approximation

$$f(x) \approx f(a) + f'(a)(x-a)$$

uses a line. Taylor polynomials extend this idea by matching higher derivatives.

Definition: Taylor polynomial

If f has n derivatives at a, the Taylor polynomial of degree n at a is

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

For example, for e^x at $a = 0$:

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}.$$

9.2 Taylor theorem with remainder

Theorem: Taylor theorem, Lagrange form of the remainder

Assume f has $n+1$ continuous derivatives near a. Then

$$f(x) = T_n(x) + R_n(x),$$

where

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$$

for some ξ between a and x.

The remainder term tells us how much error is made by truncating the Taylor series.

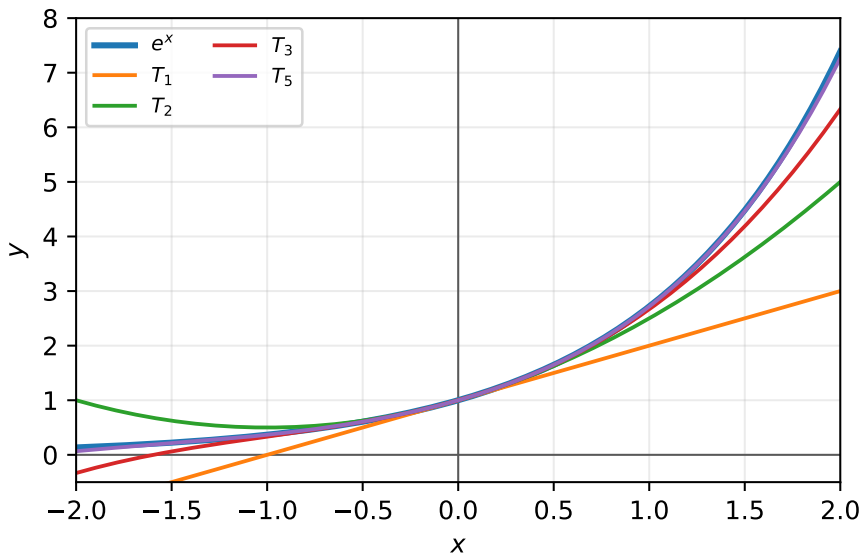

Figure 6: Taylor polynomials of increasing degree approximate e^x near the expansion point.

Computer science lens: Taylor approximation in algorithms

Taylor expansions appear in numerical methods, optimization, approximation algorithms, graphics, signal processing, and analysis of algorithms. For example, the approximation

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

explains why $\log(1+x) \approx x$ for small x.

10 One-variable optimization

10.1 Critical points

A point c is a critical point of f if either

$$f'(c) = 0$$

or $f'(c)$ does not exist.

For differentiable functions, local maxima and minima in the interior occur only at critical points.

Theorem: Fermat theorem

If f has a local maximum or local minimum at an interior point c, and f is differentiable at c, then

$$f'(c) = 0.$$

10.2 First and second derivative tests

If f' changes from positive to negative at c, then f has a local maximum at c. If f' changes from negative to positive, then f has a local minimum.

If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c. If $f''(c) < 0$, then f has a local maximum.

10.3 Newton method

Newton's method solves $f(x) = 0$ by repeatedly replacing f near x_k with its tangent line. The update is

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

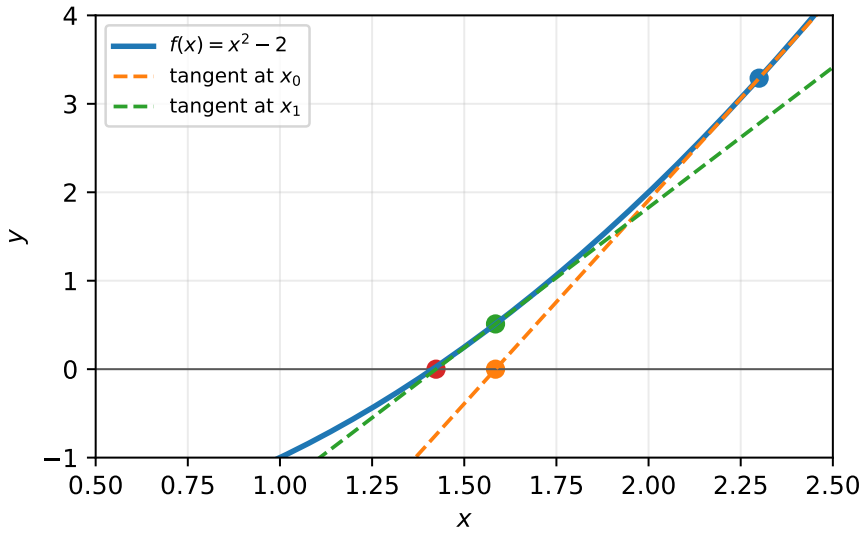

Figure 7: Newton's method uses tangent lines to generate better approximations to a root.

Listing 1: Newton iteration in mathematical pseudocode

```
choose initial point x
repeat until convergence:
    x <- x - f(x)/f'(x)
return x
```

Warning: Newton method is not automatically safe

Newton's method can converge very fast near a simple root, but it can fail if the initial point is poor, if $f'(x_k)$ is close to zero, or if the function has complicated geometry. Bisection is slower but more robust under sign-change assumptions.

11 Antiderivatives and indefinite integrals

11.1 Antiderivative

Differentiation asks for the rate of change of a function. Integration reverses this process.

Definition: antiderivative

A function F is an antiderivative of f on an interval if

$$F'(x) = f(x)$$

for all x in that interval.

The indefinite integral denotes the family of all antiderivatives:

$$\int f(x)\,\mathrm{d}x = F(x) + C.$$

The constant C is necessary because derivatives of constant functions are zero.

11.2 Basic antiderivatives

Function	Antiderivative
$x^n,\ n \neq -1$	$\frac{x^{n+1}}{n+1} + C$
$1/x$	$\log\lvert x\rvert + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\log a} + C$
$\cos x$	$\sin x + C$
$\sin x$	$-\cos x + C$

Example: a polynomial antiderivative

$$\int (3x^2 - 4x + 7)\,\mathrm{d}x = x^3 - 2x^2 + 7x + C.$$

Differentiating the result recovers the integrand.

12 Definite integrals

12.1 Accumulation as a limit of sums

The definite integral is defined as the limit of Riemann sums. Divide $[a, b]$ into n subintervals of width

$$\Delta x = \frac{b-a}{n}.$$

Choose sample points x_i^* in each subinterval. The Riemann sum is

$$\sum_{i=1}^{n} f(x_i^*)\Delta x.$$

If these sums approach a common limit as the partition becomes finer, that limit is the integral.

Definition: Riemann integral

A bounded function f on $[a, b]$ is Riemann integrable if the Riemann sums converge to the same value for all sufficiently fine partitions. The common value is denoted

$$\int_a^b f(x)\,\mathrm{d}x.$$

12.2 Signed area

For continuous functions, the integral equals signed area: area above the x-axis contributes positively, area below contributes negatively.

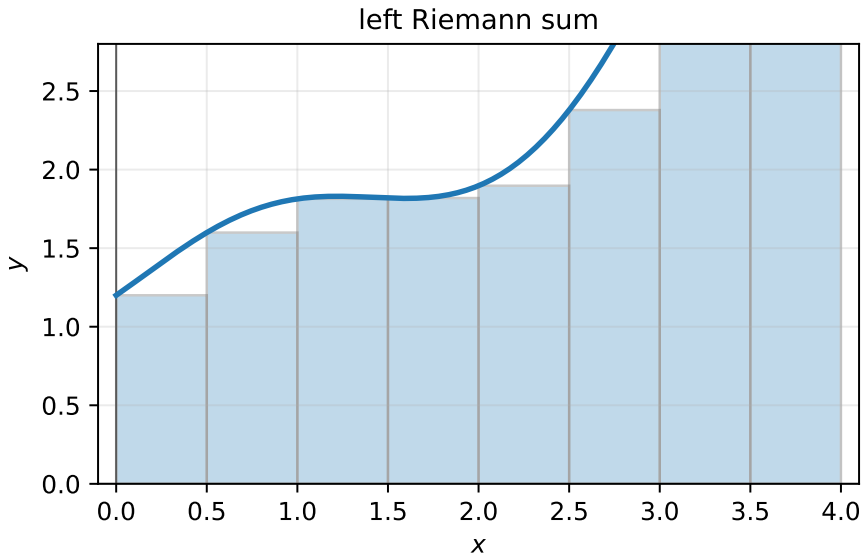

Figure 8: A Riemann sum approximates accumulated area by rectangles.

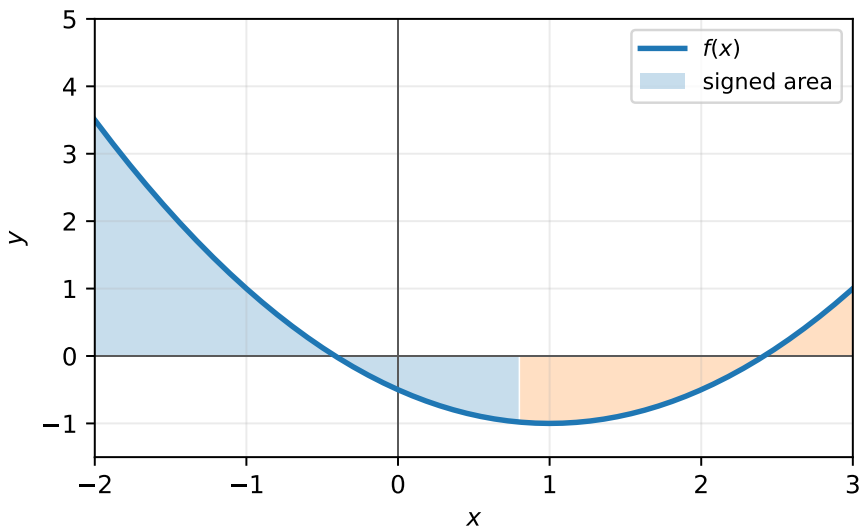

Figure 9: The definite integral represents signed accumulation.

12.3 Basic properties

For integrable functions f, g:

$$\int_a^b (f+g)\,\mathrm{d}x = \int_a^b f\,\mathrm{d}x + \int_a^b g\,\mathrm{d}x,$$

$$\int_a^b cf\,\mathrm{d}x = c\int_a^b f\,\mathrm{d}x,$$

$$\int_a^b f\,\mathrm{d}x = -\int_b^a f\,\mathrm{d}x, \qquad \int_a^a f\,\mathrm{d}x = 0.$$

If $f(x) \le g(x)$ on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x \le \int_a^b g(x)\,\mathrm{d}x.$$

13 The fundamental theorem of calculus

13.1 Part I: accumulation differentiates back

Define

$$A(x) = \int_a^x f(t)\,\mathrm{d}t.$$

If f is continuous, then

$$A'(x) = f(x).$$

Theorem: fundamental theorem of calculus, part I

If f is continuous on $[a, b]$, then the function

$$A(x) = \int_a^x f(t)\,\mathrm{d}t$$

is differentiable on (a, b), and

$$A'(x) = f(x).$$

This theorem says that integration and differentiation are inverse operations in a precise sense.

13.2 Part II: evaluating definite integrals

Theorem: fundamental theorem of calculus, part II

If $F'(x) = f(x)$ on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Example: evaluating a polynomial integral

$$\int_0^2 (3x^2 + 1)\,\mathrm{d}x = \Big[x^3 + x\Big]_0^2 = 10.$$

14 Techniques of integration

14.1 Substitution

Substitution is the integral version of the chain rule. If $u = g(x)$, then $\mathrm{d}u = g'(x)\,\mathrm{d}x$.

Theorem: substitution rule

If g is differentiable and f is continuous, then

$$\int f(g(x))g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u.$$

Example: substitution

Compute

$$\int 2x\cos(x^2)\,\mathrm{d}x.$$

Let $u = x^2$. Then $\mathrm{d}u = 2x\,\mathrm{d}x$, so

$$\int 2x\cos(x^2)\,\mathrm{d}x = \int \cos u\,\mathrm{d}u = \sin u + C = \sin(x^2) + C.$$

14.2 Integration by parts

Integration by parts comes from the product rule:

$$(uv)' = u'v + uv'.$$

Integrating both sides gives

$$\int uv'\,\mathrm{d}x = uv - \int u'v\,\mathrm{d}x.$$

Theorem: integration by parts

$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u.$$

Example: logarithm integral

Compute $\int \log x\,\mathrm{d}x$. Let

$$u = \log x, \qquad \mathrm{d}v = \mathrm{d}x.$$

Then $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$ and $v = x$. Hence

$$\int \log x\,\mathrm{d}x = x\log x - \int 1\,\mathrm{d}x = x\log x - x + C.$$

14.3 Partial fractions

Rational functions can often be integrated by decomposing them into simpler fractions. For example,

$$\frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)} = \frac{1}{2}\frac{1}{x-1} - \frac{1}{2}\frac{1}{x+1}.$$

Thus

$$\int \frac{1}{x^2-1}\,\mathrm{d}x = \frac{1}{2}\log|x-1| - \frac{1}{2}\log|x+1| + C.$$

15 Improper integrals and convergence

15.1 Infinite intervals

An improper integral over an infinite interval is defined by a limit:

$$\int_a^\infty f(x)\,\mathrm{d}x = \lim_{R\to\infty}\int_a^R f(x)\,\mathrm{d}x,$$

if the limit exists.

Example: a convergent improper integral

$$\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = \lim_{R\to\infty}\left[-\frac{1}{x}\right]_1^R = 1.$$

Example: a divergent improper integral

$$\int_1^\infty \frac{1}{x}\,\mathrm{d}x = \lim_{R\to\infty} \log R = \infty.$$

15.2 The p-test

Theorem: p-test for improper integrals

For $p > 0$,

$$\int_1^\infty \frac{1}{x^p}\,\mathrm{d}x$$

converges if and only if $p > 1$.

Computer science lens: normalization in probability and machine learning

A nonnegative function can define a probability density only if its integral over the domain is finite and can be normalized to 1. Improper integrals are therefore part of the mathematical language behind probabilistic models.

16 Numerical integration

16.1 Why numerical integration is necessary

Many integrals cannot be evaluated using elementary antiderivatives. Even when an exact formula exists, numerical integration may be preferable in computation.

16.2 Rectangle and midpoint rules

For a uniform partition of $[a, b]$ into n intervals with $\Delta x = (b - a)/n$, the left rectangle rule is

$$\int_a^b f(x)\,\mathrm{d}x \approx \Delta x \sum_{i=0}^{n-1} f(a + i\Delta x).$$

The midpoint rule is

$$\int_a^b f(x)\,\mathrm{d}x \approx \Delta x \sum_{i=0}^{n-1} f\left(a + \left(i + \frac{1}{2}\right)\Delta x\right).$$

16.3 Trapezoid rule

The trapezoid rule approximates the graph by line segments:

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{\Delta x}{2}\left(f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right).$$

Listing 2: Composite trapezoid rule

```
input: function f, interval [a,b], number of subintervals n
h <- (b-a)/n
s <- f(a) + f(b)
for i = 1,...,n-1:
    s <- s + 2*f(a+i*h)
return h*s/2
```

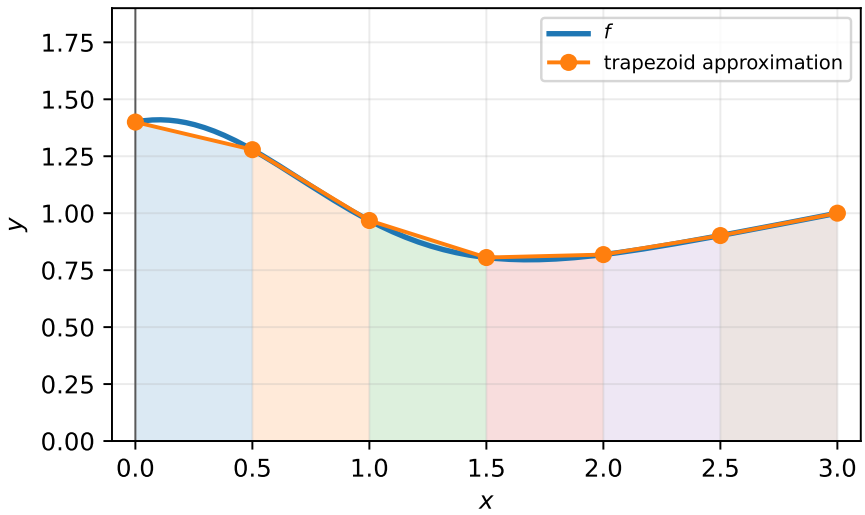

Figure 10: The trapezoid rule approximates a curve by linear pieces.

16.4 Error awareness

Numerical integration is not just computation; it is approximation. If f is sufficiently smooth, the trapezoid rule error decreases like $\mathcal{O}(h^2)$, where $h = (b-a)/n$. Simpson's rule, under stronger smoothness assumptions, has error of order $\mathcal{O}(h^4)$.

Warning: floating-point arithmetic

Making h smaller does not always improve the final result indefinitely. Rounding error and function evaluation error may eventually dominate discretization error.

17 Sequences and series

17.1 Sequences

A sequence is a function from $\mathbb{N}$ to $\mathbb{R}$:

$$n \mapsto a_n.$$

We say that $a_n \to L$ if for every $\varepsilon > 0$ there exists N such that

$$n \geq N \quad \Longrightarrow \quad |a_n - L| < \varepsilon.$$

Sequences occur naturally in iterative algorithms:

$$x_0, x_1, x_2, \ldots$$

Convergence of the algorithm means convergence of this sequence.

17.2 Series

A series is an infinite sum

$$\sum_{n=1}^{\infty} a_n.$$

It converges if the sequence of partial sums

$$s_N = \sum_{n=1}^{N} a_n$$

converges.

Example: geometric series

If $|r| < 1$, then

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

This formula appears in analysis of randomized algorithms, discounted rewards, recurrence solving, and numerical methods.

17.3 Power series

A power series has the form

$$\sum_{n=0}^{\infty} a_n (x-a)^n.$$

Taylor series are power series constructed from derivatives. They allow us to approximate functions by polynomials, which are easy for computers to evaluate.

18 Multivariable calculus

18.1 Functions of several variables

Many problems in computer science involve many variables:

$$f : \mathbb{R}^n \to \mathbb{R}.$$

Examples include loss functions depending on model parameters, image-processing energies depending on pixels, and objective functions in optimization.

18.2 Partial derivatives

The partial derivative with respect to x_i measures change in the i-th coordinate while other variables are held fixed:

$$\frac{\partial f}{\partial x_i}(x) = \lim_{h \to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(x)}{h}.$$

Example: partial derivatives

Let

$$f(x, y) = x^2 y + e^{xy}.$$

Then

$$\frac{\partial f}{\partial x} = 2xy + ye^{xy}, \qquad \frac{\partial f}{\partial y} = x^2 + xe^{xy}.$$

18.3 Gradient

Definition: gradient

For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{bmatrix}.$$

The gradient points in the direction of steepest local increase. The negative gradient points in the direction of steepest local decrease.

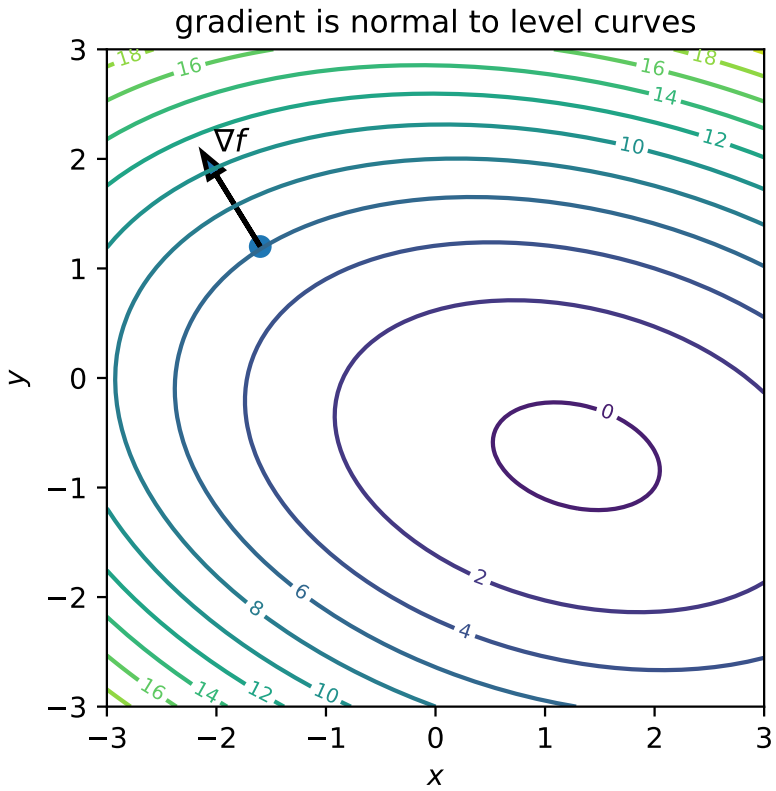

Figure 11: For a scalar function of two variables, the gradient is perpendicular to level curves.

18.4 Directional derivative

For a unit vector u, the directional derivative is

$$D_u f(x) = \lim_{h \to 0} \frac{f(x + hu) - f(x)}{h}.$$

If f is differentiable, then

$$D_u f(x) = \langle \nabla f(x), u \rangle .$$

This formula shows that the gradient determines all directional derivatives.

19 Jacobian, Hessian, and chain rule

19.1 Jacobian matrix

For a vector-valued function $F : \mathbb{R}^n \to \mathbb{R}^m$,

$$F(x) = \begin{bmatrix} F_1(x) \\ \vdots \\ F_m(x) \end{bmatrix},$$

the Jacobian matrix is

$$J_F(x) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}.$$

The Jacobian is the best linear approximation of F near x:

$$F(x+h) \approx F(x) + J_F(x)h.$$

19.2 Multivariable chain rule

If $F : \mathbb{R}^n \to \mathbb{R}^m$ and $G : \mathbb{R}^m \to \mathbb{R}^k$, then

$$J_{G \circ F}(x) = J_G(F(x))J_F(x).$$

This is the matrix form of the chain rule. It is the central formula behind automatic differentiation.

19.3 Hessian matrix

For $f : \mathbb{R}^n \to \mathbb{R}$, the Hessian is the matrix of second partial derivatives:

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

20 Gradient descent and optimization for CS

20.1 Gradient descent

To minimize $f : \mathbb{R}^n \to \mathbb{R}$, gradient descent uses the update

$$x_{k+1} = x_k - \eta \nabla f(x_k),$$

where $\eta > 0$ is the step size or learning rate.

Listing 3: Gradient descent

```
choose x and step size eta
repeat until stopping criterion:
    x <- x - eta * grad_f(x)
return x
```

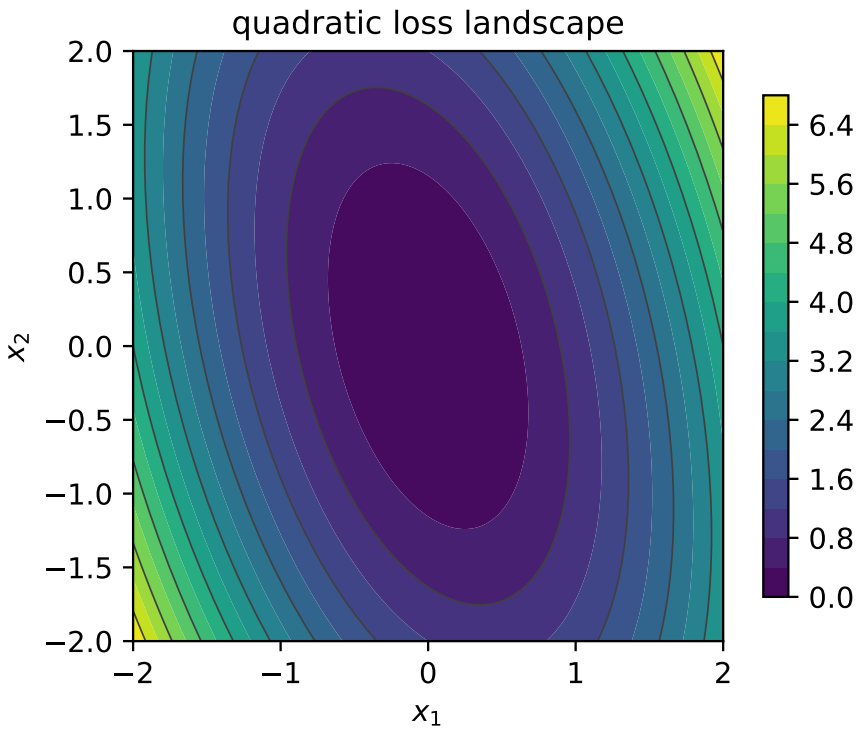

Figure 12: A quadratic objective has a constant Hessian, determining the curvature of the loss landscape.

20.2 Quadratic example

Let

$$f(x) = \frac{1}{2}x^T Ax - b^T x,$$

where A is symmetric positive definite. Then

$$\nabla f(x) = Ax - b.$$

The unique minimizer satisfies

$$Ax = b.$$

Thus solving a linear system can be viewed as minimizing a quadratic function.

Computer science lens: linear algebra meets calculus

Many numerical algorithms are combinations of linear algebra and calculus. The gradient gives a vector; the Hessian gives a matrix; optimization becomes the study of how these objects guide iterative updates.

20.3 Convex optimality condition

If f is differentiable and convex, then

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

for all x, y. Therefore, if $\nabla f(x^*) = 0$, then

$$f(y) \geq f(x^*)$$

for all y, so x^* is a global minimizer.

21 Integrals in data and algorithms

21.1 Average value

The average value of a continuous function on $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x)\,\mathrm{d}x.$$

This is a continuous analogue of the arithmetic mean.

21.2 Probability density

A probability density function p on $\mathbb{R}$ satisfies

$$p(x) \geq 0, \qquad \int_{-\infty}^{\infty} p(x)\,\mathrm{d}x = 1.$$

The expected value is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xp(x)\,\mathrm{d}x,$$

provided the integral converges.

21.3 Continuous approximation of discrete sums

Sums are sometimes approximated by integrals:

$$\sum_{k=1}^{n} f(k) \approx \int_1^n f(x)\,\mathrm{d}x.$$

For increasing positive functions, integrals provide upper and lower bounds for sums. This is useful in algorithm analysis.

Example: harmonic sum

The harmonic sum

$$H_n = \sum_{k=1}^{n} \frac{1}{k}$$

grows like $\log n$. Integral comparison gives

$$\int_1^{n+1} \frac{1}{x}\,\mathrm{d}x \leq H_n \leq 1 + \int_1^n \frac{1}{x}\,\mathrm{d}x,$$

so

$$\log(n+1) \leq H_n \leq 1 + \log n.$$

22 Common mistakes

Warning: limits are not values

A limit $\lim_{x\to a} f(x)$ can exist even if $f(a)$ is undefined or different from the limit.

Warning: continuity is not differentiability

Every differentiable function is continuous, but not every continuous function is differentiable. Corners, cusps, and vertical tangents are common sources of failure.

Warning: indefinite integrals need constants

Always include C in indefinite integrals:

$$\int f(x)\,\mathrm{d}x = F(x) + C.$$

Warning: gradient descent is local information

The negative gradient is the direction of steepest local decrease. It does not automatically guarantee global optimization unless additional assumptions, such as convexity, are satisfied.

23 Worked review examples

Example: limit and asymptotic growth

Evaluate

$$\lim_{n\to\infty} \frac{5n\log n + 2n}{n^{3/2}}.$$

We have

$$\frac{5n\log n + 2n}{n^{3/2}} = \frac{5\log n + 2}{\sqrt{n}}.$$

Since $\log n = o(\sqrt{n})$, the limit is 0. Hence

$$5n\log n + 2n = o(n^{3/2}).$$

Example: Taylor approximation

Approximate $\log(1.02)$. With $x = 0.02$,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots.$$

Using two terms,

$$\log(1.02) \approx 0.02 - \frac{0.02^2}{2} = 0.0198.$$

Example: gradient and Hessian

Let

$$f(x,y) = x^2 + xy + 2y^2 - 3x + y.$$

Then

$$\nabla f(x,y) = \begin{bmatrix} 2x + y - 3 \\ x + 4y + 1 \end{bmatrix},$$

and

$$H_f = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}.$$

The critical point solves

$$2x + y = 3, \qquad x + 4y = -1.$$

Solving gives $x = 13/7$, $y = -5/7$. Since the Hessian is positive definite, this point is a global minimizer.

Example: definite integral

Evaluate

$$\int_0^1 x e^{x^2}\,\mathrm{d}x.$$

Let $u = x^2$, so $\mathrm{d}u = 2x\,\mathrm{d}x$. Then

$$\int_0^1 x e^{x^2}\,\mathrm{d}x = \frac{1}{2}\int_0^1 e^u\,\mathrm{d}u = \frac{1}{2}(e-1).$$

24 Exercises

Limits and continuity

Exercise 24.1. Compute

$$\lim_{x\to 2} \frac{x^2-4}{x-2}.$$

Exercise 24.2. Decide whether the function

$$f(x) = \begin{cases} x^2, & x < 1, \\ 2x - 1, & x \geq 1 \end{cases}$$

is continuous at $x = 1$.

Exercise 24.3. Show that $\log n = o(n^\alpha)$ for every $\alpha > 0$.

Differentiation

Exercise 24.4. Differentiate

$$f(x) = x^3 e^{-x}.$$

Exercise 24.5. Find the tangent line to

$$f(x) = \log x$$

at $x = 1$.

Exercise 24.6. For

$$\sigma(x) = \frac{1}{1+e^{-x}},$$

prove that $\sigma'(x) = \sigma(x)(1-\sigma(x))$.

Integration

Exercise 24.7. Compute

$$\int x \log x\,\mathrm{d}x.$$

Exercise 24.8. Evaluate

$$\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x.$$

Exercise 24.9. Determine for which $p > 0$ the integral

$$\int_1^\infty \frac{1}{x^p}\,\mathrm{d}x$$

converges.

Multivariable calculus

Exercise 24.10. Compute the gradient and Hessian of

$$f(x,y) = e^{xy} + x^2 - y^2.$$

Exercise 24.11. Let

$$f(x,y) = x^2 + 4y^2.$$

Starting from $(1,1)$, write two steps of gradient descent with step size $\eta = 0.1$.

25 Short formula sheet

Limits and asymptotics

$$\lim_{x\to 0}\frac{\sin x}{x} = 1, \qquad \lim_{x\to 0}\frac{e^x-1}{x} = 1, \qquad \lim_{x\to 0}\frac{\log(1+x)}{x} = 1.$$

$$f = o(g) \iff \lim \frac{f}{g} = 0, \qquad f \sim g \iff \lim \frac{f}{g} = 1.$$

Derivatives

$$f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}.$$

$$(fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \qquad (f\circ g)' = (f'\circ g)g'.$$

$$\frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\log x = \frac{1}{x}, \quad \frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x.$$

Integrals

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a), \qquad F' = f.$$

$$\int f(g(x))g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u, \qquad \int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u.$$

Multivariable

$$\nabla f = \begin{bmatrix} \partial f/\partial x_1 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix}, \qquad D_u f(x) = \langle \nabla f(x), u\rangle.$$

$$x_{k+1} = x_k - \eta\nabla f(x_k).$$